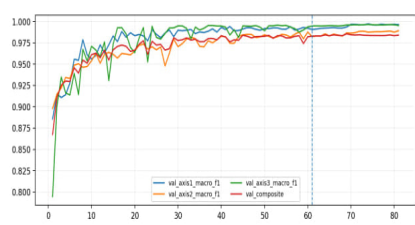
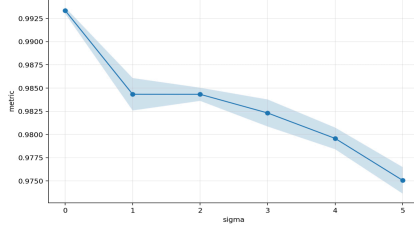
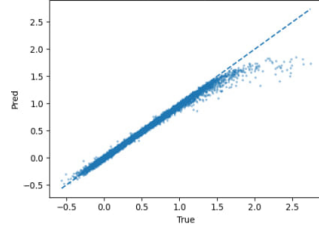


“ Project Summary ”

Project Title	Decoding Diabetic Peripheral Neuropathy Heterogeneity: A Hybrid ML System Using a Patient-Personalized Vector for Disease Representation and Stratification
Affiliation	Regeneron International Science and Engineering Fair (ISEF)
Student	Mariam Mohamed Ahmed Elelidy
Supervisor	Dr. Reda Abdelbaset Elazab Mohamed, PhD
Problem	- Diabetic peripheral neuropathy (DPN) is a common diabetes complication causing peripheral nerve damage with sensory loss, pain, and motor deficits. 1. Prevalence: • Affects ~50% of long-term diabetics, ~130 million globally; ~8–10 million adults in the U.S. 2. Prognosis & Stratification Gap: • Heterogeneity impedes personalized diagnosis; conventional tests only detect dysfunction <i>-not interpret-</i> leading to unstable, trial-and-error care. 3. Economic Burden: • Annual direct medical costs for DPN are ~\$5,000–\$25,000 per patient, totaling over \$50 billion globally, ~\$10 billion in the U.S., and ~\$0.5–1 billion in Egypt per year.
Methodology	1. Data Input • Pathophysiological mechanisms (TNF-α, MDA) capturing inflammation and oxidative stress. • EMG biomarkers (MUNE, MUI, MUAPI) reflecting motor unit loss and dysfunction. • Metadata: demographic variables, durations, clinical severity & pain score. 2. Model Architecture • Mechanism and metadata encoded via Autoencoder $\rightarrow$ <i>mechanism latent</i>. • EMG features encoded using 1D-CNN + BiLSTM $\rightarrow$ <i>EMG latent</i>. $\Rightarrow$ Latents fused using Transformer-based cHVAE to generate a Personalized Vector (PV). $\triangleright$ PV feeds: ○ Multi-head MLP classifier $\rightarrow$ stratification. ○ Multi-task MLP regressor $\rightarrow$ predictions. • Interpretability via SHAP and GNNs-based mechanisms–muscle interaction mapping. 3. System Output A) Personalized Vector (PV): patient-specific disease signature. B) Patient Stratification Summary: <i>Axis 1</i>: progression dynamics <i>Axis 2</i>: Dominant mechanism <i>Axis 3</i>: Symptom–damage alignment C) Management Risk Window D) Mechanisms–to–Muscle Interaction Map E) Nerve Biological Age F) Predicted Outcome G) Pathophysiological Fragmentation Index (PFI) H) VR-Guided Intervention (Phase II): future behavioral neuromodulation module.

Results & Analysis	Table 2: Overall Executive performance	
	Block	Metrics (Test)
	Discrimination	Accuracy/Precision/Recall $\approx$ 0.96–0.98 , F1-macro/ROC-AUC $\approx$ 0.97–0.99
	Error	MAE = 0.037 , RMSE = 0.061
	Agreement	$R^2 = 0.95$, Pearson $r = 0.96$, Spearman $\rho = 0.98$
	Calibration	ECE $\approx$ 0.002–0.006 , MCE $\approx$ 0.09–0.14
	Robustness	Δ Accuracy (SNR stress) < 3%
	Uncertainty	bootstrap CI: stable 95%
Main Figures		
		
		
Experiments	1) Augmented Generalization Cohort - N = 100,000 samples, Split: 70% train / 15% val / 15% test - Macro-F1 0.99 AUROC 0.997 Sensitivity 98.74% Specificity 99.43% MCC 0.982 Brier 0.014 Generalization gap (train–test ΔF1): 0.004 Variance CV-σ (F1): $\pm$0.003	
	2) External Human Cohort - N = 90 (60 DPN / 30 Control) “Zero retraining” - Accuracy 95.6% Macro-F1 0.954 AUROC 0.972 Sensitivity 94.8% Specificity 96.7% MCC 0.912 ECE 0.011 AUROC drop vs internal = –0.022 (<i>structural stability preserved</i>) — <i>After confirming external validity, internal structure was tested</i> — - Latent cluster separability (<i>silhouette score</i>) = 0.67 vs 0.18 (raw clinical grouping) - Hidden subtype separation effect size (<i>within DPN</i>) = Cohen’s d = 1.31 - Within-class dispersion reduction after latent encoding = –42% - Intra-DPN progression gradient correlation = r = 0.91 - Reclassification gain beyond binary clinical label = +39% - Axis monotonicity across oxidative–inflammatory gradients = 100% directional consistency	
	3) Meta-Anchor & Physiological Stress Alignment r (<i>oxidative–inflammatory concordance</i>) = 0.89 Cross-marker coherence = 0.92 - Slope amplification vs pooled literature = +24% Trend concordance gain = +39% - Effect size (model stratification vs severity gradients) = Cohen’s d = 1.42 - Nonlinear fit R² = 0.93 Hazard-direction agreement = 87% Directional inversion = 0.002% - Local slope (<i>k-region</i>) amplification factor: $\times$2.4 vs baseline zone	
Impact	Table 2: Performance estimates derived from internal cross-validation;	
	Key Benefit	Value
	Reduced Trial-and-Error	↓ 60%
	Lower Risk of Complications	↓ 35%
	Reduced Treatment Cost	↓ 45%
	Personalization Level	↑ 95–98%
	Reliable Decision Support	↑ 92% consistency
	Early Risk Identification	↑ 73%
	Disease Re-Interpretation	Model-driven shift in disease understanding